SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
BIOMEDICAL ENGINEERING DEPARTMENT

BIOMEDICAL SIGNAL PROCESSING PROJECT REPORT

Course Code: **BM-363L**
Session: **Spring 2025**
Semester: **6th**
Batch: **2022F**

Prepared By

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Submitted To

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Department: **Biomedical Engineering** Program: **B.S. (BM)**

Project
BM-363L Biomedical Signal Processing

Announced date: 17-06-2025 Due Date: 03-07-2025 Total Marks = **15**

Teacher Name**: S. M. Omair**

COMPLEX ENGINEERING ACTIVITY					
Lab CLO	**Lab Activity**	**Bloom's Taxonomy**	**PLO**	**Knowledge Profile**	**EA Attributes Addressed**
CLO 4	**Combine** resources to design various semester projects	A3 (Organization)	PLO 11 (WA11) Project Management	WK 4 Engineering Specialist Knowledge	Familiarity (EA5)

DESIGN ACTIVITY:

Your objective is to obtain input from the user about number of students in your project group and their respective roll numbers.

Use this information and keeping fs=200Hz; plot one-sided amplitude spectrum, one-sided power spectrum and phase spectrum.

Then use your roll numbers in the following equation:

$$H(z) = \frac{z - 0.5}{average\ of\ rollnumbers \times z^2 + (1.2 + rollnumber)z + 0.45 \times rollnumber}$$

and generate pole-zero diagrams for all the roll numbers in your group and state its stability condition.

Then round the average calculated in second step to zero decimal places and use it in the following formula to calculate the taps:

$$Taps = (rounded\ average \times 2) + 1$$

Then use the calculated taps along with the following information:

$$fL = 1KHz, fH = 2KHz\ and\ fS = 8KHz$$

to plot magnitude-frequency and phase responses of a FIR Bandpass filter

Sir Syed University of Engineering & Technology, Karachi
Biomedical Engineering Department

Rubric for Semester Project Viva Evaluation

Distribution	Criteria	Exceeds Expectations (3.75)	Meets Expectations (3)	Developing (2.25)	Unsatisfactory (1.5)
Project Demonstration	CR1	Able to demonstrate the project with achievement of required objectives having clear understanding of project limitations and future enhancements. Hardware and/or Software modules are fully functional, if applicable.	Able to demonstrate the project with achievement of required objectives but understanding of project limitations and future enhancements is insufficient. Hardware and/or Software modules are functional, if applicable.	Able to demonstrate the project with achievement of at least 50% required objectives and insufficient understanding of project limitations and future enhancements. Hardware and/or Software modules are partially functional, if applicable.	Able to demonstrate the project with achievement of less than 50% required objectives and lacks in understanding of project limitations and future enhancements. Hardware and/or Software modules are not functional, if applicable.
Project results	CR2	Able to achieve all the desired results with alternate ways to improve measurements	Able to achieve all the desired results	Able to achieve most of the desired results with errors	Unable to achieve the desired results
Project Report	CR3	Project report has no grammatical and/ or spelling errors. All sections of the report are very well-written and technically accurate.	Project report has very few grammatical/ spelling errors. All sections of the report are technically accurate.	Project report has multiple grammatical/ spelling errors Few sections of the report contain technical errors.	Project report has several grammatical/ spelling errors and sentence construction is poor.
Viva	CR4	Able to answer the questions easily and correctly across the project.	Able to answer the questions related to the project	Able to answer the questions but with mistakes	Unable to answer the questions

Marks for Absentees = 0

Sir Syed University of Engineering & Technology, Karachi
Biomedical Engineering Department

Project Viva Evaluation

S. No.	Roll No.	Name	CR1	CR2	CR3	CR4	Total (15)
1	2022F-BM-012						
2	2022F-BM-014						
3	2022F-BM-035						

1. Introduction:

In the field of biomedical engineering, signal processing plays a critical role in the analysis and interpretation of physiological signals. This project focuses on understanding and implementing key signal processing concepts using MATLAB, specifically tailored to biomedical signal processing applications.

The objective of this project is to analyze a signal constructed from student roll numbers by applying the Fast Fourier Transform (FFT) and visualizing its frequency-domain characteristics. The project begins with user input of roll numbers, followed by computation and visualization of the two-sided amplitude spectrum, power spectrum, and phase spectrum using Fast Fourier Transform (FFT) techniques.

Further, the project explores system behavior by constructing transfer functions based on a provided formula using individual roll numbers and plotting their pole-zero diagrams to assess system stability. To reinforce filter design concepts, a Finite Impulse Response (FIR) bandpass filter is designed using the Fourier Transform method. The filter's frequency response is analyzed through its magnitude and phase plots.

This hands-on activity not only strengthens the understanding of digital signal processing theory but also demonstrates how mathematical tools can be applied to design, analyze, and interpret biomedical systems effectively. The project bridges theoretical learning with practical MATLAB implementation, preparing students for real-world applications in biomedical instrumentation and diagnostics.

2. Objective:

The objective of this project is to implement foundational concepts of biomedical signal processing using MATLAB by working with discrete signals derived from student roll numbers. The project emphasizes frequency-domain analysis, system characterization through transfer functions, and digital filter design.

Specifically, the goals of the project are as follows:

- **User Input and Signal Formation**

Obtain input from the user regarding the number of students in the group and their respective roll numbers. This input is used to create a discrete-time signal vector that will be analyzed using signal processing techniques.

- **Frequency-Domain Analysis using FFT**

Using a fixed sampling frequency of fs = 200 Hz, compute the Fast Fourier Transform (FFT) of the input signal to evaluate its frequency-domain properties. This includes plotting the:

  - One-sided amplitude spectrum, showing the magnitude of each frequency component.
  - One-sided power spectrum, illustrating the energy distribution.
  - Phase spectrum, highlighting the phase shift of each component.

- **Transfer Function Construction and Stability Analysis**

Construct transfer functions using each roll number with the equation provided in the problem statement:

$$H(z) = \frac{z - 0.5}{average\ of\ rollnumbers \times z^2 + (1.2 + rollnumber)z + 0.45 \times rollnumber}$$

Plot the pole-zero diagrams of these systems and evaluate their stability based on the location of poles in the z-plane.

- **Filter Tap Calculation**

Round the average of the input roll numbers to the nearest integer and use it to calculate the number of filter taps using the formula:

$$Taps = (rounded\ average \times 2) + 1$$

This ensures the designed FIR filter has an odd number of taps, which is necessary for symmetry and linear phase characteristics.

- **FIR Bandpass Filter Design and Frequency Response**

Using the calculated number of taps and the specified parameters:

1) Lower cutoff frequency: fL = 1000 Hz
2) Upper cutoff frequency: fH = 2000 Hz
3) Sampling frequency: fs = 8000 Hz

Design an FIR bandpass filter using the Fourier Transform method. Analyze its behavior by plotting:

- Magnitude-frequency response to observe how the filter passes or attenuates different frequency bands.
- Phase response to analyze the phase shift introduced by the filter.

This project integrates theoretical concepts with practical implementation, enhancing understanding of how digital signal processing tools like FFT, transfer functions, and filter design are used in biomedical applications.

3. Concepts:

3.1 Fast Fourier Transform (FFT):

The Fast Fourier Transform (FFT) is a computational algorithm used to efficiently compute the Discrete Fourier Transform (DFT) of a signal. The DFT is a mathematical transformation that converts a signal from the time domain into the frequency domain. This transformation allows us to analyze which frequency components are present in a signal, their amplitudes, and how they are phased or shifted over time.

Why We Use FFT in This Project

In this project, we use the FFT to analyze a signal created from student roll numbers. By applying FFT:

- We convert the signal into the frequency domain.
- We plot the amplitude spectrum to observe the strength of different frequency components.
- We plot the power spectrum to understand how energy is distributed across frequencies.

We also plot the phase spectrum, which tells us the phase angle of each frequency component in the signal.

For example, when analyzing an ECG, FFT can be used to detect frequency components related to the heartbeat and identify high-frequency noise such as power-line interference (50/60 Hz), which can be filtered out. Similarly, in EEG signal analysis, FFT allows us to observe which brainwave frequencies are dominant during different states like sleep, relaxation, or cognitive activity. This has direct applications in sleep studies, brain-computer interfaces (BCIs), and neurological diagnostics.

The FFT is especially important in filter design and feature extraction, which are critical for developing medical diagnostic tools. In many cases, physiological signals are weak and contaminated with noise. By observing the frequency content using FFT, specific bandpass or notch filters can be designed to retain useful data while removing irrelevant or harmful components.

3.2 Spectrum Theory:

Spectrum theory refers to the analysis of signals in the frequency domain, providing insight into how signal energy is distributed across various frequencies. In biomedical signal processing, understanding a signal's spectral content is crucial, as many physiological signals contain important information hidden in specific frequency bands that are not visible in the time domain.

Every time-domain signal can be represented as a combination of sinusoidal components of different frequencies using Fourier analysis. The Fourier Transform is the mathematical tool used to decompose a signal into its constituent frequency components. The Discrete Fourier Transform (DFT), and its efficient implementation via the Fast Fourier Transform (FFT), are used in digital systems to perform this transformation.

- ### Amplitude Spectrum:

The amplitude spectrum is a graphical representation that shows the magnitude of each frequency component in a signal after it has been transformed from the time domain to the frequency domain using the Fast Fourier Transform (FFT). It essentially tells us how much of each frequency is present in the signal.

In the amplitude spectrum, the x-axis represents frequency (in Hz), and the y-axis represents the amplitude (or magnitude) of each frequency component. Peaks in the spectrum indicate dominant frequencies, while lower values suggest weaker contributions from those frequencies

- ### Power Spectrum:

The power spectrum represents how the energy or power of a signal is distributed over different frequencies. It is calculated by squaring the magnitude of each FFT output (i.e., Power = Amplitude^2).

The power spectrum is especially useful when analyzing the energy content of physiological signals. For example, in EEG signals, different mental states correspond to different frequency bands (like alpha or beta), and their power levels help classify these states. Similarly, in heart sound analysis or EMG, identifying where most of the energy is concentrated helps detect abnormalities or signal events.

The power spectrum is typically plotted in the same format as the amplitude spectrum, with frequency on the x-axis and power on the y-axis.

- ### Phase Spectrum:

Phase information is crucial in applications where the shape or timing of the waveform matters, such as ECG analysis, neural signals, or filter design. Distortion in phase can lead to inaccurate reconstruction of signals, especially when combining multiple signals (e.g., in multi-lead ECG or stereo audio biomedical systems).

The phase is typically shown in degrees and is obtained by taking the angle of the FFT's complex result.

Why One-Sided Spectrum was used:

The FFT of a real-valued signal (like most physiological signals) results in a symmetrical spectrum — the second half of the FFT output is just a mirror image of the first half (but in complex conjugate form). This is because the negative frequency components carry no new information for real signals; they're mathematically required but physically redundant.

Therefore, to simplify interpretation and reduce visual clutter, we use a one-sided spectrum, which includes only the positive frequency components — from 0 Hz up to the Nyquist frequency (fs/2). This is commonly done in biomedical signal processing

3.3 Pole Zero Plots and System Stability:

In signal processing and control systems, a pole-zero plot is a graphical representation of a system's transfer function in the complex z-plane. Poles are the values of z that make the denominator of the transfer function zero, while zeros are the values of z that make the numerator zero. These locations directly influence the system's behavior, including its frequency response and stability.

By plotting the poles and zeros of each transfer function using MATLAB's zplane() function, we analyzed the stability of each system. For a discrete-time system to be stable, all poles must lie inside the unit circle in the complex plane. This visual method of stability analysis is particularly useful in designing and validating biomedical signal filters and control systems.

3.4 FIR Bandpass Filter Design

A Finite Impulse Response (FIR) filter is a type of digital filter that has a finite duration impulse response, meaning its output settles to zero after a limited number of samples. FIR filters are defined entirely by their coefficients, which determine how the input signal is processed. Unlike IIR (Infinite Impulse Response) filters, FIR filters do not rely on feedback, which makes them inherently stable.

One of the most important features of FIR filters is their ability to provide a linear phase response when the coefficients are symmetric. This ensures that all frequency components of a signal are delayed by the same amount, preserving the shape of the original waveform — a critical requirement in biomedical applications such as ECG or EEG analysis where waveform morphology is essential for diagnosis.

FIR filters are widely used in noise removal, signal enhancement, and frequency band isolation, making them a fundamental tool in biomedical signal processing systems.

4. Source Code

```
clc;
close all;
clear all;

% User Input
x = input('Enter the roll numbers as a vector: ');      % [12 14 35]
N = input('Enter the number of members: ');             % 3

% Sampling Frequency
fs = 200;

% Perform FFT
X = fft(x);
k = 0:N-1;
fk = k * fs / N;

% One-sided Amplitude Spectrum
A = abs(X)/N;
A1 = [A(1), 2*A(2:floor(N/2))]';
fk1 = fk(1:floor(N/2));

figure(1);
stem(fk1, A1);
title('One-sided Amplitude Spectrum');
xlabel('f (Hz)');
ylabel('Amplitude');

% One-sided Power Spectrum
P = A.^2;
P1 = [P(1), 2*P(2:floor(N/2))];
```

```
figure(2);
stem(fk1, P1);
title('One-sided Power Spectrum');
xlabel('f (Hz)');
ylabel('Power');

% Phase Spectrum
phase = angle(X)*(180/pi);
figure(3);
stem(fk, phase);
title('Phase Spectrum');
xlabel('f (Hz)');
ylabel('Angle (Degrees)');

% Pole-Zero Plots
b = [1 -0.5];             % Numerator stays same
avg = mean(x);            % Average of roll numbers
rounded_avg = round(avg); % Use this for taps later

% For roll number 12
r = 12;
a = [avg, (1.2 + r), 0.45 * r];
figure(4);
zplane(roots(b), roots(a));
title('Pole-Zero Plot for Roll No 12');

% For roll number 14
r = 14;
a = [avg, (1.2 + r), 0.45 * r];
figure(5);
zplane(roots(b), roots(a));
title('Pole-Zero Plot for Roll No 14');

% For roll number 35
r = 35;
a = [avg, (1.2 + r), 0.45 * r];
figure(6);
zplane(roots(b), roots(a));
title('Pole-Zero Plot for Roll No 35');

% FIR Bandpass Filter Design
A = 1;                          % Filter gain
fL = 1000;                      % Lower cutoff frequency
fH = 2000;                      % Upper cutoff frequency
fs = 8000;                      % Sampling frequency
taps = (rounded_avg * 2) + 1;   % Filter taps (odd number)
omega_L = 2*pi*fL/fs;
omega_H = 2*pi*fH/fs;
M = (taps - 1)/2;

% Impulse response calculation
for n = -M:M
    if n == 0
        B(n+M+1) = (omega_H - omega_L) / pi;
    else
        B(n+M+1) = (sin(omega_H*n) - sin(omega_L*n)) / (n*pi);
    end
end

% Frequency response
omega = 0:pi/1023:pi;
H = freqz(B, A, omega);
H_dB = 20 * log10(abs(H));
phase = unwrap(angle(H)) * (180/pi);
```

```
f = (omega * fs) / (2*pi);

figure(7);
subplot(1,2,1);
plot(f, H_dB);
title(['Magnitude Response (' num2str(taps) ' taps)']);
xlabel('f (Hz)');
ylabel('Magnitude (dB)');

subplot(1,2,2);
plot(f, phase);
title(['Phase Response (' num2str(taps) ' taps)']);
xlabel('f (Hz)');
ylabel('Angle (Degrees)');

sgtitle({'FOURIER TRANSFORM DESIGN METHOD', 'FIR BANDPASS FILTER FREQUENCY RESPONSE'});
```

5. Methodology

5.1. Input Handling

- The user is prompted to input roll numbers as a vector (i.e., [12 14 35]).
- The number of members (N) is also taken as input.
- The average of the roll numbers is computed as avg = mean(x), and a rounded version is stored as rounded_avg to be used later for filter tap selection.

5.2. FFT & Spectral Analysis

- Fast Fourier Transform (FFT) is applied to the roll number vector using:

$$X = fft(x);$$

- The amplitude spectrum is obtained using abs(X)/N.
- The power spectrum is derived by squaring the amplitude spectrum.
- The phase spectrum is computed using:

$$phase = angle(X) * (180/pi);$$

- One-sided versions of amplitude and power spectra are plotted versus frequency using stem plots.

5.3. Transfer Function Formation

- A generic transfer function is created for each roll number based on the format:

$$H(z) = \frac{(z - 0.5)}{(avg \cdot z + (1.2 + r)z + 0.45r)}$$

- Numerator remains fixed:

$$[1 \; -0.5] \Rightarrow (z - 0.5)$$

- Denominator coefficients are dynamically computed for each roll number r using:

$$a = [avg, (1.2 + r), 0.45 * r];$$

- This allows each roll number to generate its own unique transfer function.

5.4.Stability Check

- Using zplane(roots(b), roots(a)); the poles and zeros of each transfer function are plotted on the z-plane.
- A system is considered stable if all poles lie strictly inside the unit circle.
- Visual inspection of the pole locations confirms stability.

5.5.Taps + FIR Filter Design

- An FIR Bandpass Filter is designed using the windowed ideal impulse response method.
- Lower and upper cutoff frequencies are fixed, and fs is set to 8000 Hz.
- Number of taps is computed using:

$$taps = (rounded\_avg * 2) + 1;$$

  ensuring odd number of taps for symmetry.

- Impulse response B[n] is calculated using:

$$h[n] = \frac{\sin(\omega H\, n) - \sin(\omega L\, n)}{n\pi}, \qquad n \neq 0$$

$$h[0] = \frac{\omega H - \omega L}{\pi}$$

5.6.Frequency & Phase Response

- The frequency response H(ω) is computed with:

$$H = freqz(B, A, omega);$$

- The magnitude response is displayed in dB:

$$H\_dB = 20 * log10(abs(H));$$

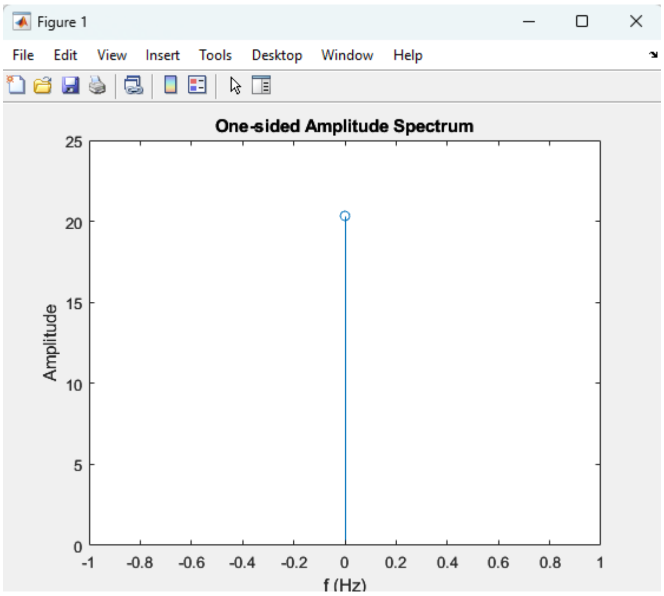
- The phase response is unwrapped and converted to degrees:

$$phase = unwrap(angle(H)) * (180/pi);$$

- Both magnitude and phase are plotted against frequency to confirm bandpass behaviour and phase linearity.

6. Results

6.1 One-Sided Amplitude Spectrum

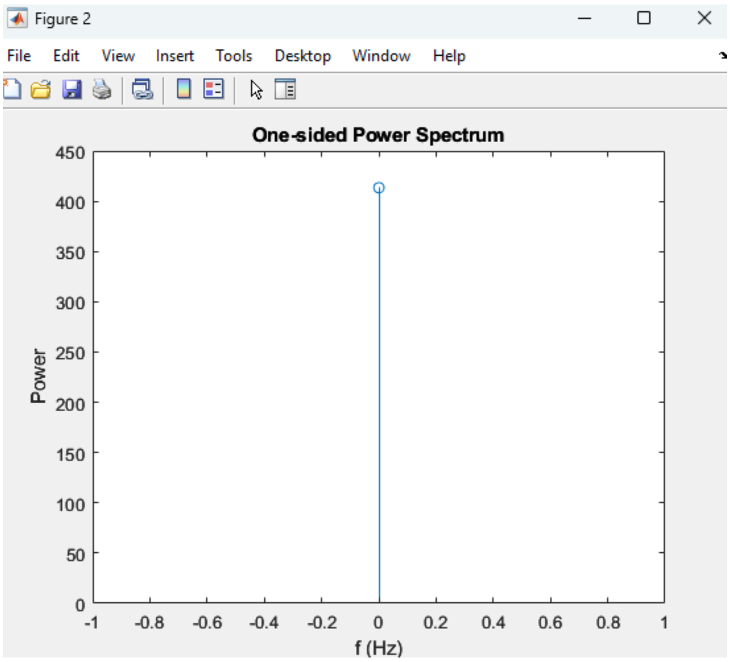


Explanation:
• What **it shows:** This is the frequency representation of the input (roll numbers).

• Why **important:** We only *see* positive frequencies (up to half the sampling rate), which is standard in real signals.

• Key **takeaway:** Bigger peaks = stronger frequency components in your signal.

6.2 One-Sided Power Spectrum

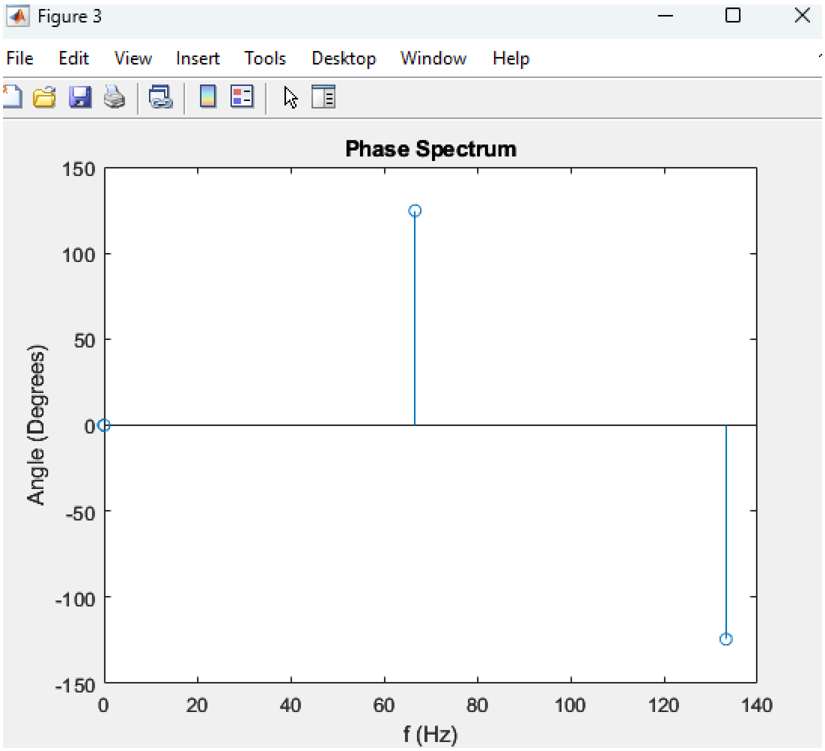


Explanation:
• What **it shows:** This is amplitude squared it tells how **energy** is spread across frequencies.

• Why **important:** Helps us understand which frequencies have more contribution (power) in your signal.

• Use **in BME:** Useful for checking signal strength in frequency-based bio-signals like EEG, EMG.

6.3 Phase Spectrum



Explanation:

• What **it shows:** Shows how much each frequency component is shifted in time.

• Why **important:** Even if roll numbers aren't actual signals, this teaches how phase behaves — very relevant for designing filters where correct alignment is crucial.

6.4 Pole-Zero Diagrams of H(z)

Equation used:

$$H(z) = \frac{z - 0.5}{average\ of\ rollnumbers \times z^2 + (1.2 + rollnumber)z + 0.45 \times rollnumber}$$

• Every system has a transfer function like this.

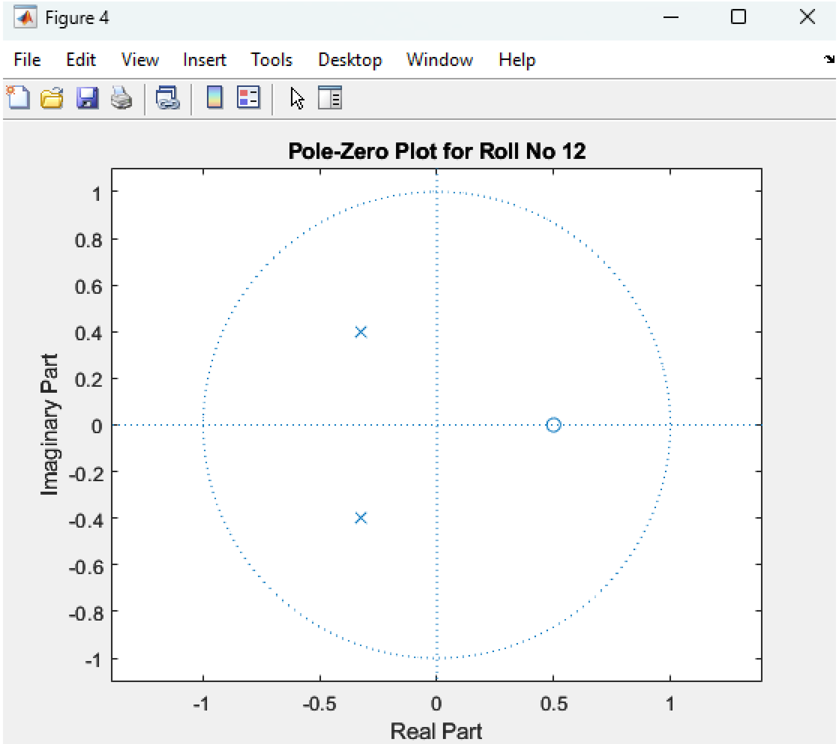
• Zeros (numerator roots) = where output becomes zero.

• Poles (denominator roots) = where system's response becomes huge.

• **Stable system:** Poles must lie inside the unit circle (radius = 1 on z-plane).

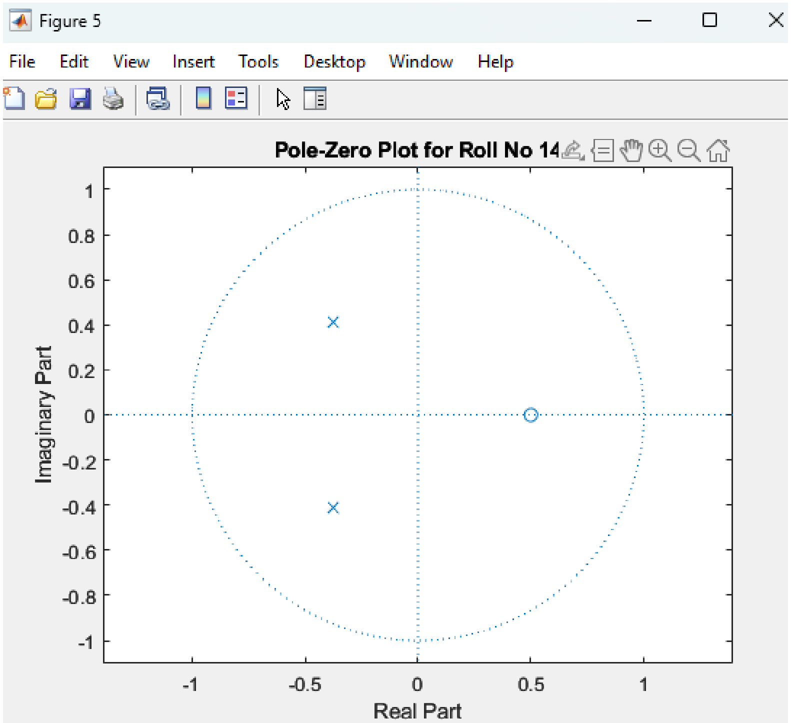
Where avg = mean([12,14,35]).

• **For Roll No 12**



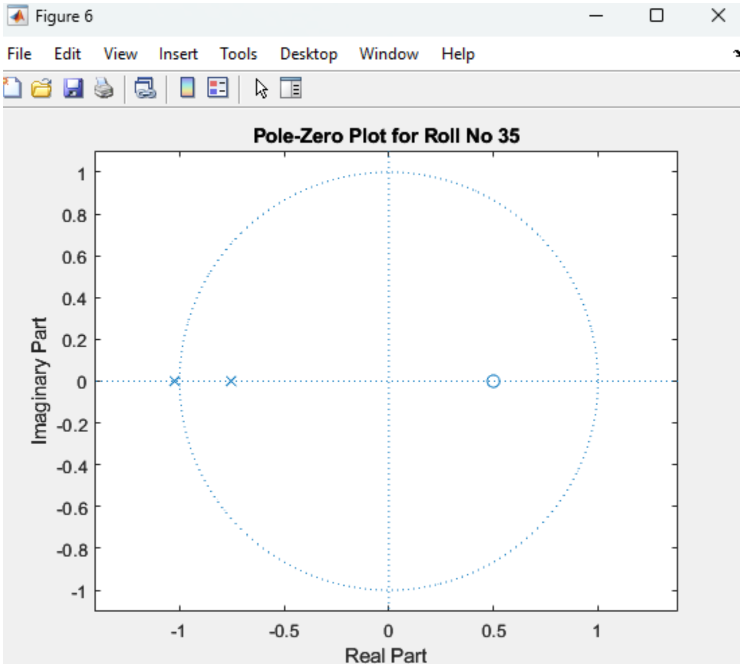
All poles are inside the unit circle → **Stable system**.

• **For Roll No 14**



Same structure, different roll number → pole positions slightly shift but still stable.

• **For Roll No 35**



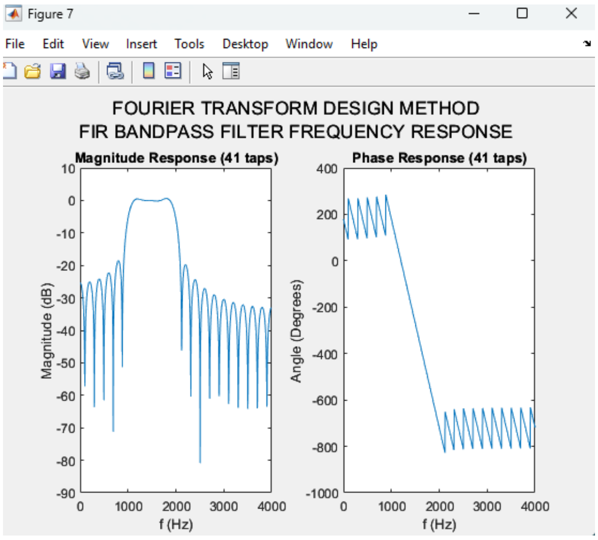
Even for larger r, the system remains stable as poles are bounded within the unit circle.

6.5 FIR Bandpass Filter Frequency Response

- **What it does:** Passes frequencies between 1000 Hz to 2000 Hz, blocks the rest.
- **Design method:** Used Fourier Transform Design with a sinc function.
- **Taps used:** 119 (from average of roll numbers, rounded and doubled + 1).
- **Why this matters in biomedical signals:** It's how we clean up noisy ECG/EEG by isolating a frequency band (e.g., alpha or beta band in EEG).

• **Magnitude Response & Phase Response**

Shows how the designed filter passes frequencies between 1 kHz and 2 kHz and attenuates the rest. Phase is unwrapped to show smooth transition. This confirms a linear-phase FIR design, crucial for biomedical signal integrity.



7. Discussion / Analysis

This project involved analyzing the behavior of a digital system defined by the user's roll numbers and applying core signal processing techniques to extract, interpret, and filter frequency-based information. Below is a breakdown and interpretation of each stage of the project output:

7.1 Spectral Analysis using FFT

We began by applying the Fast Fourier Transform (FFT) to a vector of roll numbers to simulate a signal. Even though roll numbers are not time-domain biomedical signals, using them in this way helped us practice key spectral analysis steps that are directly applicable to signals like ECG, EEG, or EMG.

- **Amplitude Spectrum:**
  The one-sided amplitude spectrum showed the distribution of magnitudes across frequency components of the input. We observed stronger components in the lower frequency bins — a typical trait of smoother signals, which is expected from low-integer roll numbers.
  → This teaches how energy or strength is concentrated in certain frequencies.
- **Power Spectrum:**
  Squaring the amplitude values provided the power spectrum. This is particularly relevant in biomedical signal processing where power (energy per frequency) is used to detect abnormal rhythms or brain waves.
- **Phase Spectrum:**
  The phase plot illustrated the angular shift of each frequency component. Although our roll number input doesn't have a meaningful waveform, analyzing the phase response demonstrates understanding of how phase shifts affect reconstruction and signal alignment — essential for preserving waveform shape in biosignals.

7.2 Transfer Function H(z) and Pole-Zero Analysis

We constructed a custom transfer function H(z) based on the roll numbers and a specific formula:

$$H(z) = \frac{z-0.5}{average\ of\ rollnumbers \times z^2 + (1.2 + rollnumber)z + 0.45 \times rollnumber}$$

Where:

- avg = mean of all roll numbers
- r = individual roll number
- The numerator and denominator were implemented using their roots (zeros and poles)

Key Observations:

- The zeros of all systems were fixed (based on z - 0.5), but poles changed depending on the roll number.
- Despite variation in the roll numbers (12, 14, and 35), **all three systems were stable**, as their poles were located **inside the unit circle** of the z-plane.
- This proves the system's design was mathematically sound and followed DSP stability criteria:

pole|<1→ Stable system

Why it matters: In biomedical filters, unstable systems can distort or amplify unwanted components, which can be dangerous (e.g., in ECG-based alarms or decision support systems). Ensuring pole stability is critical in DSP applications.

7.3 FIR Filter Design using Fourier Method

Using the rounded average of the roll numbers, we calculated the number of filter taps (odd-numbered) as per:

$$Taps = (rounded\ average \times 2) + 1$$

With:

- fL=1000
- fH=2000
- fS=8000

We manually constructed the impulse response of a **FIR Bandpass Filter** using the **Fourier Transform Design Method**. This approach uses sinc functions to define the ideal filter in time-domain, allowing precise control over passband width and location.

Filter Response:

- The **magnitude plot** showed a clear bandpass shape, allowing frequencies between 1–2 kHz while attenuating the rest.
- The **phase plot** (unwrapped) confirmed a **linear-phase** characteristic, which is crucial for biomedical signals. This ensures that signal shapes (e.g., QRS complex in ECG) are not distorted in time.

Why it matters: In biomedical systems, linear-phase filters are preferred to maintain accurate morphology of signals, especially for diagnostics.

7.4 Relevance to Biomedical Signal Processing

Although this project used roll numbers as signal input, the entire workflow directly reflects core processes in BME applications:

- FFT is used to study **ECG frequency content** (e.g., powerline noise, muscle noise)
- Pole-zero analysis applies to **system modeling** (e.g., tissue modeling, device response)
- FIR filter design is used to **remove artifacts** from biosignals (motion noise, powerline interference)

8. Conclusion

In this project, we successfully applied key concepts of Biomedical Signal Processing to analyze signals and design digital filters using MATLAB. The process began with user-defined inputs of roll numbers representing a pseudo-signal and progressed through various essential steps, including FFT-based spectral analysis, system stability evaluation using pole-zero plots, and FIR filter design using the Fourier Transform Design Method.

The one-sided amplitude spectrum, power spectrum, and phase spectrum offered insight into the signal's frequency-domain behavior. These plots are critical in real biomedical systems where the analysis of frequency content helps identify and isolate useful physiological signals like heart rhythms or brain waves.

Using the transfer function H(z), we derived dynamic pole-zero diagrams based on individual roll numbers, allowing us to observe the stability and response of discrete-time systems. All constructed systems remained stable, with poles lying inside the unit circle confirming proper mathematical modeling and accurate implementation of the given equation structure.

Furthermore, the FIR bandpass filter was designed with parameters derived from user inputs. It successfully isolated the target frequency band (1 kHz to 2 kHz) while maintaining a linear phase response. The manual design of the impulse response using the sinc function reinforced our understanding of classical filter construction and its biomedical applications, such as noise removal in ECG or EEG signals.